4 ACCELERATION DUE TO GRAVITY USING SIMPLE PENDULUM

AIM To measure the acceleration due to gravity using a simple pendulum

APPARATUS REQUIRED Retort stand, pendulum bob, thread, meter scale, stop watch.

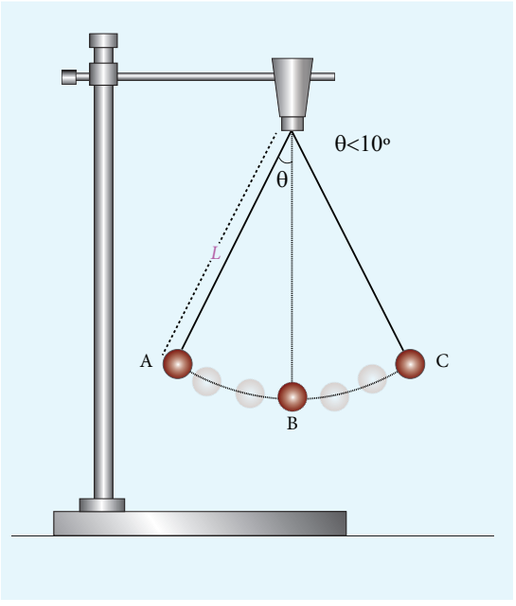
FORMULA Acceleration due to gravity $g = 4\pi^2\left(\frac{L}{T^2}\right)$ (m s^{-2})

where $T \leftarrow$ Time period of simple pendulum (second)

$g \leftarrow$ Acceleration due to gravity (metre sec^{-2})

$L \leftarrow$ Length of the pendulum (metre)

DIAGRAM



PROCEDURE

- Attach a small brass bob to the thread.
- Fix this thread on to the stand.
- Measure the length of the pendulum from top of the suspension hook to the middle of the bob of the pendulum. Record the length of the pendulum in the table given below.
- Note down the time (t) taken for 10 oscillations using stop watch.
- The period of oscillation $T = \frac{t}{10}$ is calculated.
- Repeat the experiment for different lengths of the pendulum 'L'. Find acceleration due to gravity g using the given formula.

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